

Literature Survey Machine Learning for Vehicle Fuel Efficiency Prediction Using the Auto MPG Dataset

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1 Introduction

Predicting vehicle fuel efficiency, measured in Miles Per Gallon (MPG), is important for environmental sustainability, vehicle design, and consumer decision-making. The Auto MPG dataset from the UCI Machine Learning Repository is a widely used benchmark for this problem. It contains vehicle attributes such as engine displacement, horsepower, weight, acceleration, model year, and origin. This survey reviews key studies that apply machine learning and neural network techniques to fuel efficiency prediction.

2 The Auto MPG Dataset

The Auto MPG dataset contains approximately 398 vehicle records from the 1970s and 1980s. Features include cylinders, displacement, horsepower, weight, acceleration, model year, and origin, while MPG serves as the target variable. The dataset is widely used because it provides a realistic regression problem with both linear and nonlinear relationships among variables.

3 Neural Network Classification

Aliyu, Adeshina Siraj (2014)

This study approached fuel efficiency prediction as a classification problem by grouping vehicles into efficiency categories. A Multi-Layer Perceptron (MLP) trained using backpropagation was used for prediction. The study showed that an MLP could successfully classify vehicle fuel efficiency using the Auto MPG dataset. It also found that removing outliers during preprocessing helped improve the model's performance and generalisation.

4 Machine Learning Across Historical and Modern Datasets

Sarkar Hasan (2025)

This study compared machine learning models using both the classic Auto MPG dataset and a modern vehicle dataset. The authors investigated how the factors affecting fuel efficiency have changed over time and evaluated multiple machine learning algorithms across different periods. The study demonstrated that vehicle characteristics remain strong predictors of fuel efficiency while highlighting changes brought by advances in vehicle technology.

5 CNN and XGBoost for Fuel Efficiency and Emission Prediction

Çeken Gemci (2025)

This study combined the Auto MPG dataset with the EPA Fuel Economy dataset containing over 38,000 vehicle records. Several algorithms were evaluated, including Random Forest, Support Vector Machines, Artificial Neural Networks, CNNs, and XGBoost. CNN achieved strong classification performance, while XGBoost produced the best regression results. The study demonstrated the effectiveness of combining deep learning and ensemble learning approaches for transportation applications.

6 Comparison of Related Studies

Table 1: Comparison of Related Studies

Study	Year	Dataset(s)	Task	Best Method
2014	UCI Auto MPG	Classification	MLP heightSarkar	Hasan
2025	UCI Auto MPG + Modern Data	Both	Multiple ML Models heightÇeken	Gemci
2025	UCI Auto MPG + EPA Dataset	Both	CNN / XGBoost height	

7 Research Gaps

Several challenges remain in this area. First, most studies do not include modern hybrid or electric vehicles. Second, model explainability remains limited, making it difficult to understand why predictions are made. Third, most models are evaluated on static datasets rather than real-time vehicle data.

8 Conclusion

Machine learning techniques have demonstrated strong performance for vehicle fuel efficiency prediction. Early studies relied primarily on MLPs, while more recent research has expanded to ensemble methods and deep learning architectures such as CNNs. Current evidence suggests that advanced machine learning models can effectively predict fuel efficiency, although opportunities remain for improving explainability and extending models to modern vehicle technologies.

References

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